

The background is a gradient of deep blue and purple, speckled with white dots resembling a starry sky. Overlaid on this are several faint, white, circular and semi-circular patterns. Some of these patterns have tick marks and numbers, suggesting a circular scale or a clock face. The numbers visible include 140, 150, 160, 170, 180, 190, 200, 210, 220, 230, 240, 250, and 260. There are also curved arrows indicating a direction of movement or flow.

LEAD SCORE CASE STUDY

Problem Statement

X Education specializes in selling online courses to industry professionals, promoting its offerings across multiple websites and search engines, including Google.

Upon visiting the website, individuals have the option to explore available courses, complete course-related forms, or view instructional videos. When they choose to provide their email addresses or phone numbers via these forms, they are categorized as leads. Additionally, the company acquires leads through referrals from previous customers or sources.

Once these leads are obtained, the sales team initiates contact by making phone calls, sending emails, and employing various methods. During this process, some of the leads successfully convert into paying customers, while the majority do not. It's worth noting that the typical lead conversion rate at X Education hovers around 30%.

Business Goal

X Education requires assistance in identifying and prioritizing the most promising leads, those with the highest likelihood of converting into paying customers.

The company is seeking a model that assigns a lead score to each of the leads, with the objective of prioritizing customers. Higher lead scores will indicate a greater chance of conversion, while lower lead scores will suggest a lower likelihood of conversion. This model will aid in focusing resources and efforts on leads with the greatest potential for conversion.

The CEO has set an approximate target lead conversion rate of 80% as the desired goal for the company.

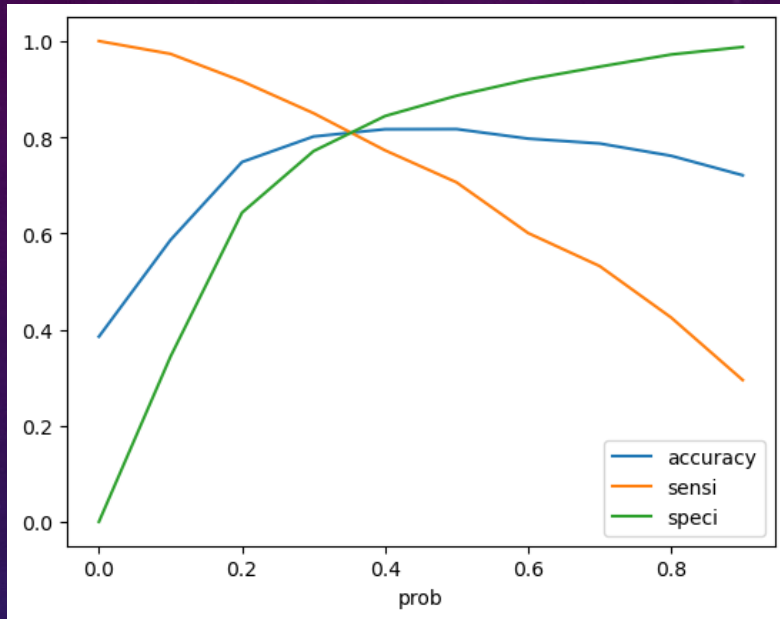
Strategy

- ❖ Acquire the data for analysis.
- ❖ Clean and preprocess the data.
- ❖ Conduct Exploratory Data Analysis (EDA).
- ❖ Standardize the features (Feature Scaling).
- ❖ Divide the data into Training and Testing datasets.
- ❖ Construct a Logistic Regression model to compute Lead Scores.
- ❖ Assess the model's performance using various metrics like Sensitivity and Specificity or Precision and Recall.
- ❖ Apply the best-performing model to the Test data based on Sensitivity and Specificity metrics.

Variables Impacting the Conversion Rate

- ❖ Do Not Email
- ❖ Total Time Spent On Website
- ❖ Lead Origin – Landing Page Submission
- ❖ Lead Source – Olark Chat
- ❖ Lead Source – Reference
- ❖ Lead Source – Welingak Website
- ❖ Last Activity – Olark Chat Conversation
- ❖ Last Activity – Other Activity
- ❖ Last Activity – SMS Sent
- ❖ Specialization – Others
- ❖ Working Professional
- ❖ Last Notable Activity

Model Evaluation - Sensitivity and Specificity on Train Dataset



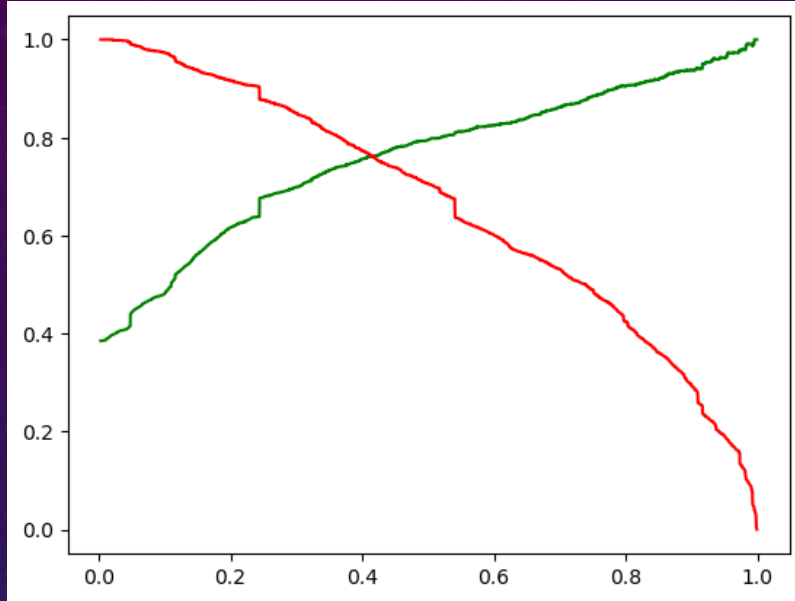
The chart illustrates an optimal cutoff value of 0.37, determined by considering factors such as Accuracy, Sensitivity, and Specificity.

Confusion Matrix

3151	754
447	1999

- ❖ Accuracy – 81.09%
- ❖ Sensitivity – 81.73%
- ❖ Specificity – 80.69%
- ❖ False Positive Rate – 19.30%
- ❖ Positive Predictive Value – 72.61%
- ❖ Negative Predictive Value – 87.58%

Model Evaluation- Precision and Recall on Train Dataset



Confusion Matrix

3461	444
719	1727

❖ Precision – 79.54%

❖ Recall – 70.60%

The graph reveals an optimal cutoff value of 0.42, determined with a focus on Precision and Recall.

Model Evaluation – Sensitivity and Specificity on Test Dataset

Confusion Matrix

1396	338
193	796

- ❖ Accuracy – 80.50%
- ❖ Sensitivity – 80.49%
- ❖ Specificity – 80.51%

Conclusion

- ❖ Although we've assessed both Sensitivity-Specificity and Precision-Recall metrics, the final prediction is determined using the optimal cutoff based on Sensitivity and Specificity.
- ❖ The test set exhibits Accuracy, Sensitivity, and Specificity values of approximately 81%, 80%, and 81%, respectively, which are closely aligned with the corresponding values computed using the training set. Additionally, the calculated lead scores provide further insights into the data.